Code Generations and Optimization - SDI - 2025

In collaboration with
IBM **SkillsBuild**

Student Transcript

Student Developer Initiative Program

Profile

Student Name : Muhamad Bayu Fadayan
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Course : Code Generations and Optimization

About Program

Student Developer Initiative is a collaborative program between IBM and Hacktiv8 aimed at equipping students and young talents with practical skills in data, programming, and AI. The main goal of this program is to help participants leverage artificial intelligence (AI), particularly IBM Granite, to enhance their productivity and support their activities—especially in software development.

Skill/Competencies

Lesson	Objective	Learning Hour
Use Generative AI for Software Development	• Participants are able to write code using IBM Granite. • Participants are able to identify common issues in AI-generated code. • Participants are able to troubleshoot common code-related issues. • Participants are able to identify appropriate models for code generation & programming tasks.	3 Hours 4 JP
Code Generation and Optimization using IBM Granite Models	• Participants are able to prompt IBM Granite models effectively. • Participants are able to apply structured workflows for reviewing and optimizing AI-generated code.	3 Hours 4 JP
Project Building	• Participants will be able to create a website or application with the help of AI (IBM Granite) to accelerate, improve, and document the code development process.	6 Hours 8 JP
Total Learning Hour		**12 Hours / 16 JP**

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Project Score

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Aspect	Description	Weight	Score
Background & Objective	Participant's ability to demonstrate an understanding of real-world problems or specific needs, clearly and logically explain project objectives, link the solution to user benefits or real-world context, and show clear cause-and-effect thinking.	**15%**	**76.09**
Technologies Used	Participant's ability to explain the core technologies (languages, frameworks, tools) and their functions, demonstrate why these technologies were chosen, show the relationship between technology and project requirements (fit-for-purpose), and ensure alignment with the project's objectives.	**25%**	**75.00**
Features	Participant's ability to explain features that are relevant to the project objectives, show the user flow and logical function, interconnect features to form a coherent user experience (end-to-end flow), and demonstrate problem-solving for users through the features.	**30%**	**72.86**
AI Support Explanation	Participant's ability to explain the role of AI in the development process (e.g., assisting in ideation, debugging, optimization, documentation, etc.), demonstrate understanding of how and why AI is used, reflect on the real-world impact (efficiency, quality, new insights), and show awareness of the learning process with AI use.	**30%**	**77.86**
Final Score	**75.38**		

Hosted by Hacktiv8 Indonesia

Riza Fahmi
Co-Founder & Chief Evangelist of Hacktiv8

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